

	00	01	11	10	
00	0	1	3	2	}
01	4	5	7	6	
11	12	13	15	14	
10	8	9	11	10	

	000	001	011	010	110	111	101	100	
00	0	1	3	2	6	7	5	4	}
01	8	9	11	10	14	15	13	12	
11	24	25	27	26	30	31	29	28	}
10	16	17	19	18	22	23	21	20	

		$\overline{C}$		C	
		00	01	11	10
$\overline{A}$ A	00	0	1	3	2
	01	4	5	7	6
	11	12	13	15	14
	10	8	9	11	10
		$\overline{D}$	D		$\overline{D}$

Diagram illustrating a 4x4 Karnaugh map (K-map) for a 4-variable function. The variables are A, B, C, and D. The map is organized with A and B as row and column indices, and C and D as the internal variables.

The map is divided into four quadrants by dashed lines:

- Top-left quadrant (A=0, B=0): Labeled $\overline{A}$ and $\overline{B}$.
- Top-right quadrant (A=0, B=1): Labeled $\overline{A}$ and B .
- Bottom-left quadrant (A=1, B=0): Labeled A and $\overline{B}$.
- Bottom-right quadrant (A=1, B=1): Labeled A and B .

The map shows the following values in the cells:

- Row 0 (A=0): 0, 1, 3, 2
- Row 1 (A=1): 4, 5, 7, 6
- Row 2 (A=2): 12, 13, 15, 14
- Row 3 (A=3): 8, 9, 11, 10

The map is also labeled with the variables $\overline{C}$, C , $\overline{D}$, and D along the top and bottom edges.

Diagram illustrating a 3-variable Karnaugh map (K-map) for variables A, B, and C. The map is a 4x4 grid with rows labeled by A and B, and columns labeled by D and E. The map is partitioned into groups of 1s and 0s.

Variables and Groupings:

- A:** {000, 001, 011, 010, 110, 111, 101, 100}
- B:** {000, 001, 011, 010, 110, 111, 101, 100}
- C:** {000, 001, 011, 010, 110, 111, 101, 100}
- D:** {000, 001, 011, 010, 110, 111, 101, 100}
- E:** {000, 001, 011, 010, 110, 111, 101, 100}
- F:** {000, 001, 011, 010, 110, 111, 101, 100}

Map Data (1s and 0s):

Row (A, B)	000	001	011	010	110	111	101	100
000	1	0	1	2	6	7	1	0
001	1	1	11	10	14	15	1	12
011	0	1	27	26	30	31	29	28
010	16	17	19	18	22	23	21	20
110	1	48	49	51	50	54	53	52
111	56	57	59	58	62	63	61	60
101	40	41	43	42	46	47	45	44
100	32	33	35	34	38	39	37	36

Groupings:

- Group 1 (Orange):** 1s at (000, 001), (001, 001), (000, 011), (001, 011).
- Group 2 (Blue):** 1s at (000, 000), (001, 000), (000, 011), (001, 011).
- Group 3 (Blue):** 1s at (000, 101), (001, 101), (000, 100), (001, 100).